THULASIRAMAN A T

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EDUCATION

Vellore Institute of Technology **Vellore, Tamil Nadu**

Integrated M.Tech in Software Engineering *August 2023 – Present*

- **Concentrations:** Software Engineering, Artificial Intelligence, Design
- **CGPA:** 8.69/10
- **Related Coursework:** Data Structures & Algorithms, Objects & Design, Computer Organization & Programming, Artificial Intelligence, Object-Oriented Programming, Agile Development

PROJECTS & EXPERIENCE

Smart India Hackathon ***September 2025***

Kochi Metro Optimization System | Team Metrix

- Developed an AI-based decision-support system for metro train scheduling using predictive modeling to optimize train deployment and reduce idle time.
- Built a predictive maintenance module using machine learning techniques to identify potential system failures, improving operational reliability.
- Designed role-based dashboards for administrators and operators, enabling secure access control and real-time monitoring.
- Addressed scheduling inefficiencies by modeling constraints such as passenger load, peak hours, and resource allocation, improving system efficiency.

Networks Information Security ***October 2025***

Password Security Analyzer | Full Stack Developer (Team) - ***website***

- Developed a full-stack password security platform using Flask, HTML, CSS, and JavaScript for real-time strength analysis.
- Implemented entropy-based password evaluation and pattern detection to assess resistance against brute-force attacks.
- Integrated breach detection APIs (HaveIBeenPwned) to identify compromised credentials in real-time.
- Designed a secure password generator with customizable complexity, improving password robustness for end users.

Machine Learning & Data Mining ***October 2025***

Solar Energy Forecasting System | Machine Learning Engineer (Team) - ***link***

- Built a machine learning model to forecast solar power generation using 300K+ historical and real-time data points.
- Applied regression-based models (e.g., Linear Regression / Random Forest / XGBoost) for accurate power prediction.
- Performed feature engineering and normalization to improve model performance and stability across varying conditions.

SKILLS

- **Language Proficiency:** English, Tamil, Telugu
- **Programming & Frameworks:** Python, SQL, JavaScript
- **Design & Tools:** Canva, Photoshop, Illustrator, MS word, VS Code, GitHub, StarUML, Jenkins (Basic),

STRENGTHS

Rivera'26 Fest – Sales Team Coordinator (VIT VELLORE)

- Engaged with 100+ (students and faculty) to pitch and sales initiatives, strengthening outreach and conversion efforts.
- Built and managed professional relationships while operating in a fast-paced environment, improving communication, adaptability, and team collaboration.

ACHIEVEMENTS & CERTIFICATIONS

- Designed the official logo for IPR & TT Cell, VIT Vellore.
- Created event posters and visual creatives for various clubs at VIT